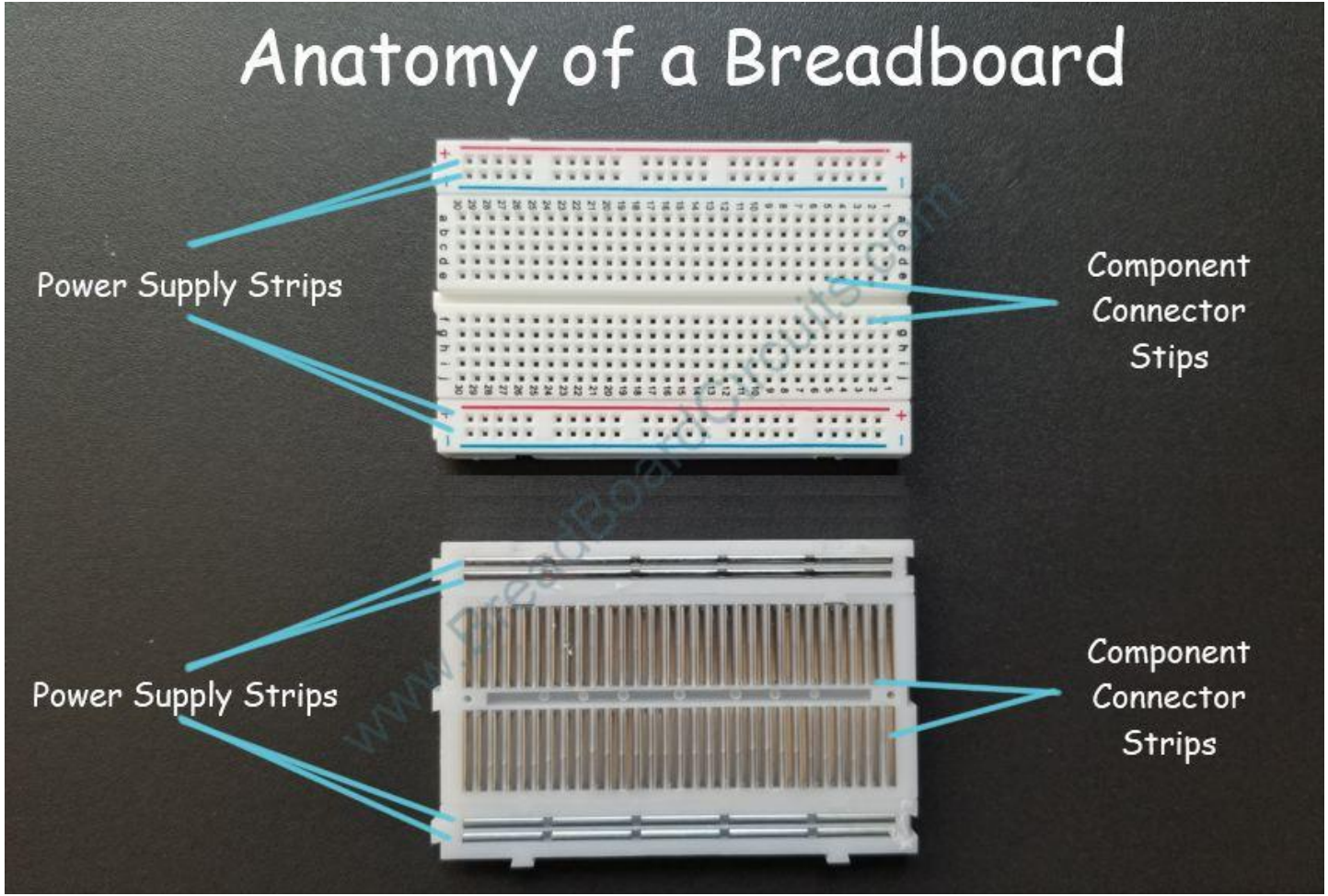
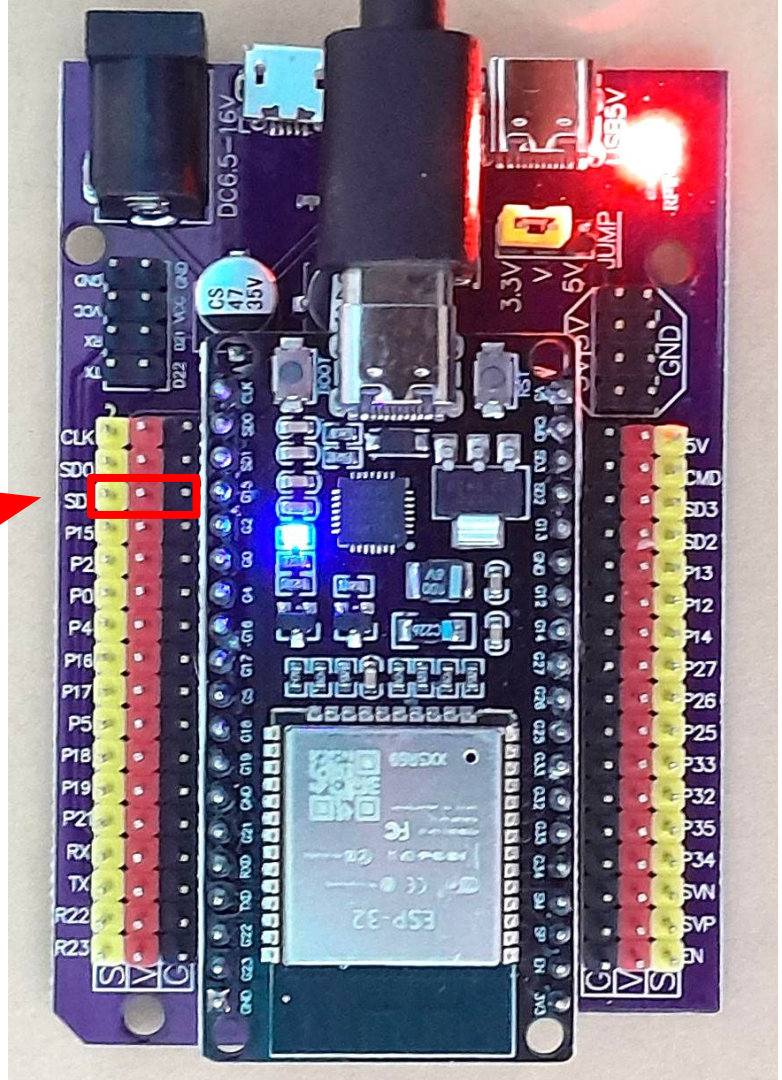


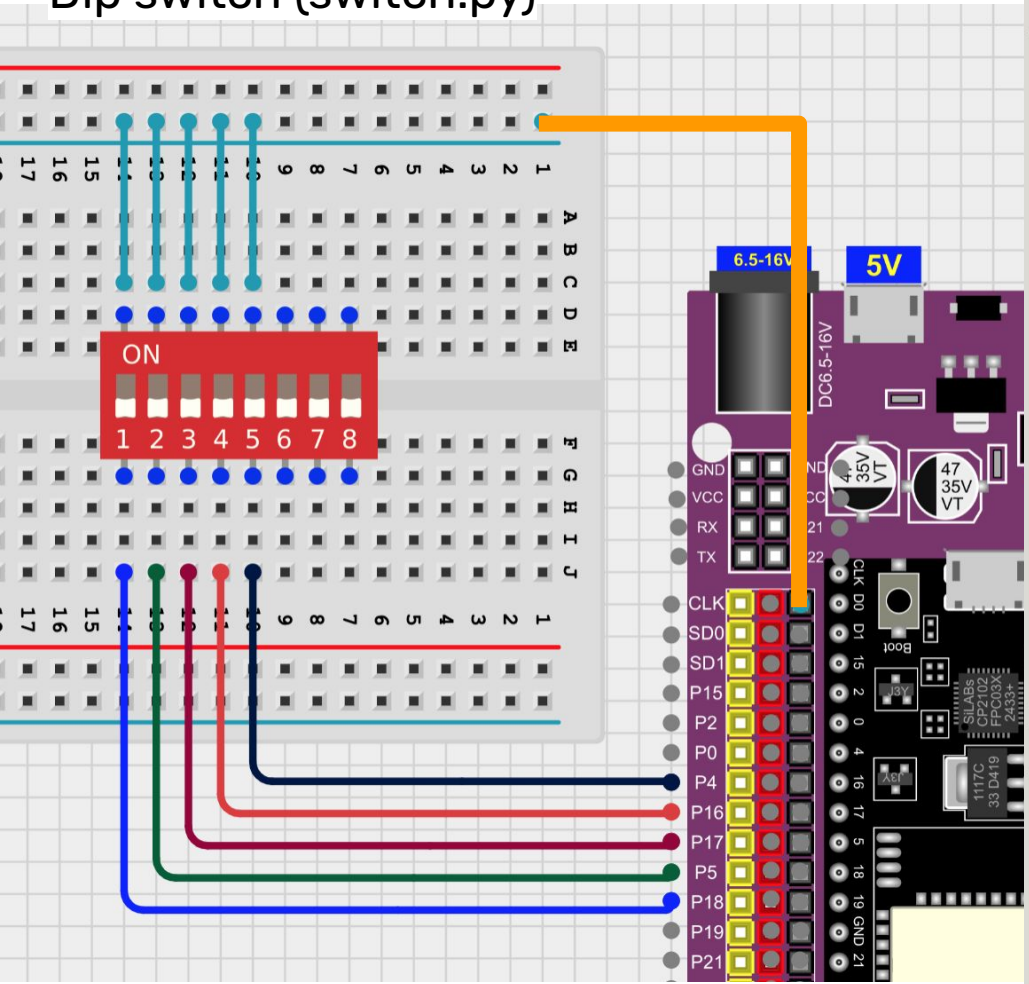
Src: https://www.breadboardcircuits.com

Esp32 breakout board

- Presents GPIO pins as Signal(yellow), Power(red) and Ground(Black)
- Most sensors and devices can be connected in this 3-pin configuration



Dip switch (switch.py)



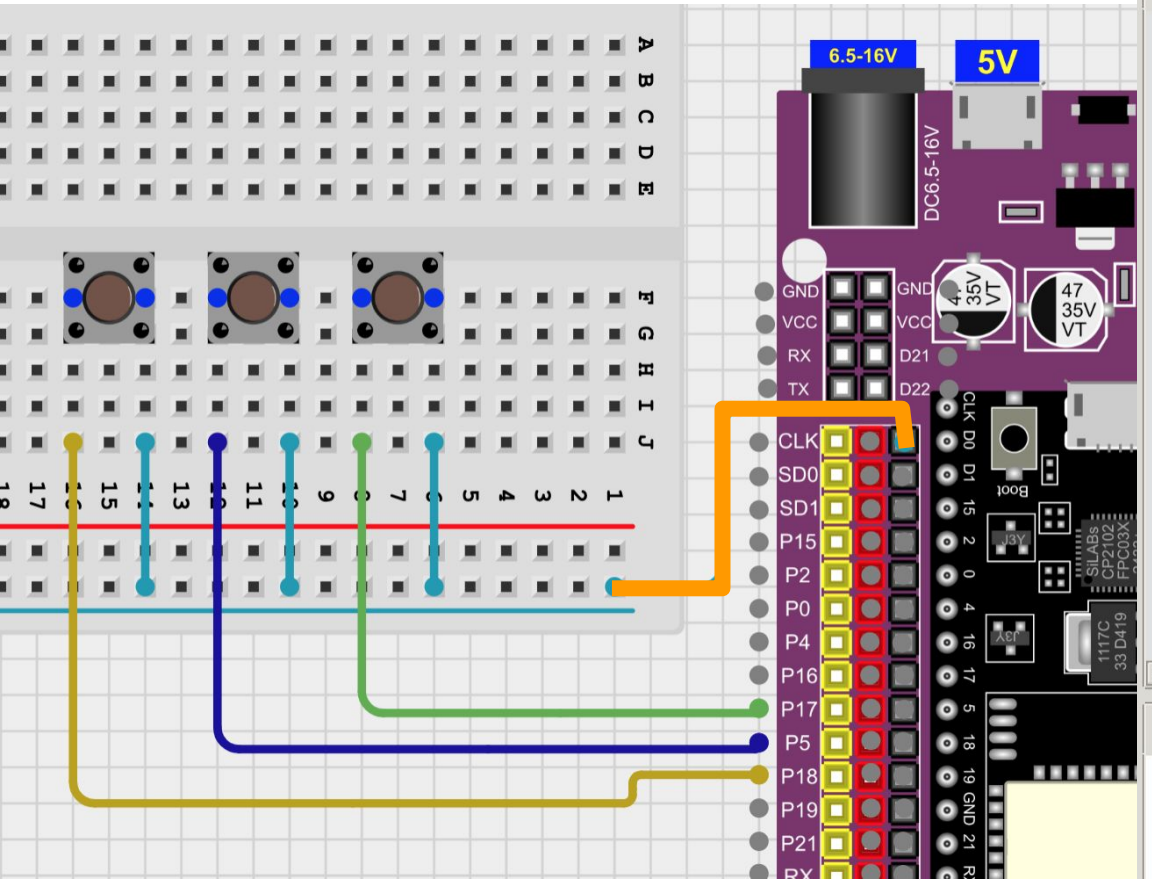
switch.py

```
1  from machine import Pin
2  import time
3
4  # Define the number of positions and the GPIO pins for ea
5
6  SWITCH_PINS = [18, 5, 17, 16, 4]  # Update these based or
7  POSITION_NUM = len(SWITCH_PINS)
8
9  ON = 0
10 OFF = 1
11
12 switches = []
13 for pin_num in SWITCH_PINS:
14     switches.append(Pin(pin_num, Pin.IN, Pin.PULL_UP))
15
16 print("DIP Switch Reader Started...")
17
18 try:
19     while True:
20         # Read and print the state of each switch positio
21         for i in range(POSITION_NUM):
22             state = switches[i].value()
23             if state == ON :
```

Shell

```
Position 3: OFF
Position 4: ON
Position 5: ON

Position 1: OFF
Position 2: ON
Position 3: OFF
Position 4: ON
Position 5: ON
```

Push buttons (switch.py)



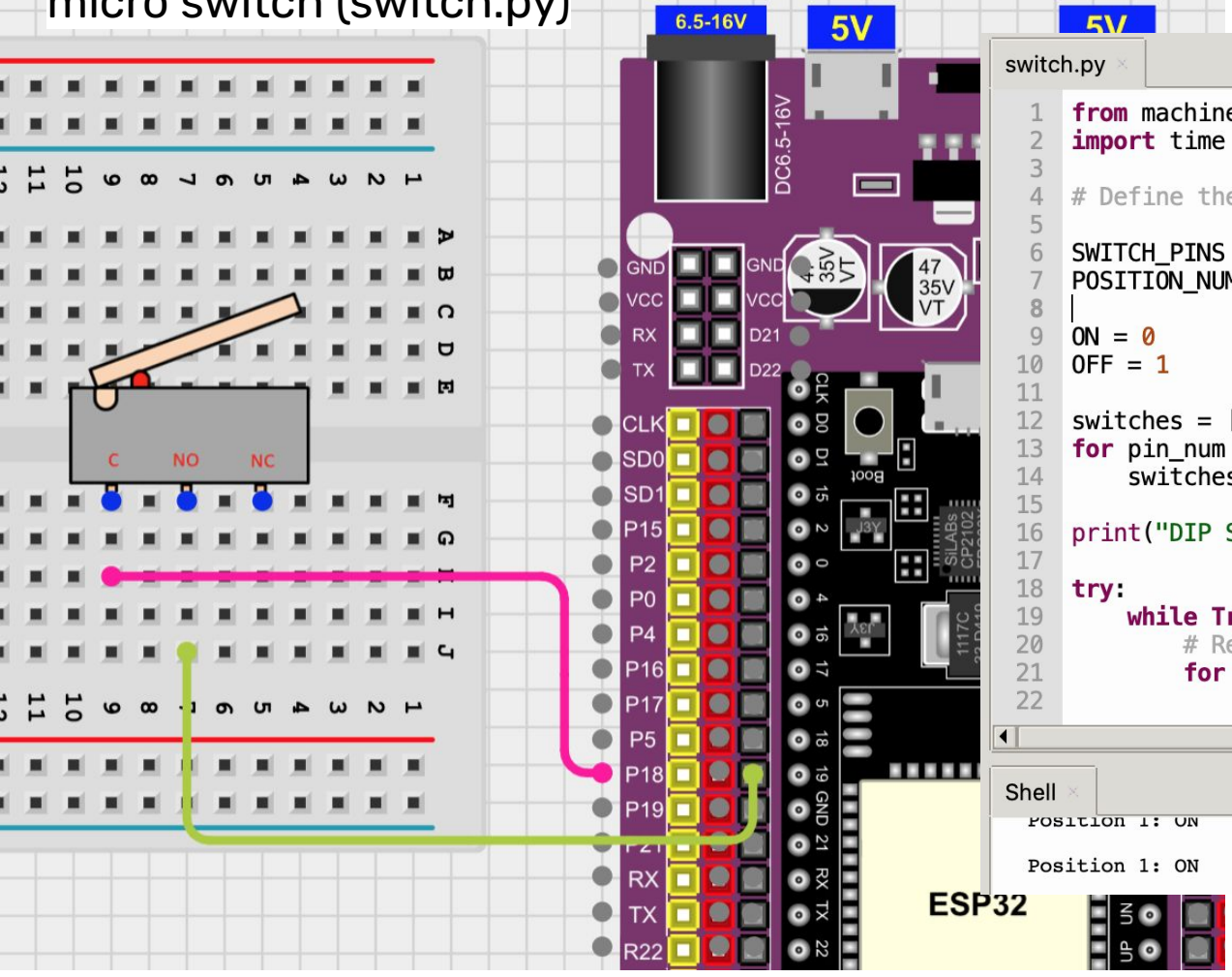
switch.py

```
1  from machine import Pin
2  import time
3
4  # Define the number of positions and the GPIO pi
5
6  SWITCH_PINS = [18, 5, 17]  # Update these based
7  POSITION_NUM = len(SWITCH_PINS)
8
9  ON = 0
10 OFF = 1
11
12 switches = []
13 for pin_num in SWITCH_PINS:
14     switches.append(Pin(pin_num, Pin.IN, Pin.PUL
15
16 print("DIP Switch Reader Started...")
17
18 try:
19     while True:
20         # Read and print the state of each switc
21         for i in range(POSITION_NUM):
22             state = switches[i].value()
```

Shell

```
Position 3: OFF

Position 1: OFF
Position 2: OFF
Position 3: OFF
```

micro switch (switch.py)



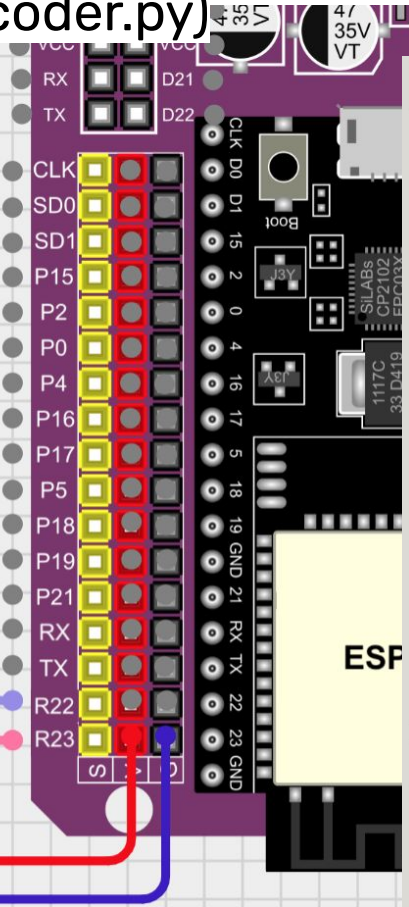
switch.py

```python
from machine import Pin
import time

# Define the number of positions and the GPIO pins for e

SWITCH_PINS = [18]  # Update these based on your wiring
POSITION_NUM = len(SWITCH_PINS)

ON = 0
OFF = 1

switches = []
for pin_num in SWITCH_PINS:
    switches.append(Pin(pin_num, Pin.IN, Pin.PULL_UP))

print("DIP Switch Reader Started...")

try:
    while True:
        # Read and print the state of each switch positi
        for i in range(POSITION_NUM):
            state = switches[i].value()
```

Shell

```
Position 1: ON

Position 1: ON
```

Rotary encoder (rotary_encoder.py)



rotary_encoder.py

```
1  from machine import Pin
2  from rotary_irq_esp import RotaryIRQ
3
4  # Initialize the rotary encoder with internal pull-
5  rotary_encoder = RotaryIRQ(
6      pin_num_clk=22,    # connect to CLK
7      pin_num_dt=23,     # connect to DT
8      pull_up=True       # Enable internal pull-up res
9  )
10
11 # Optional: set boundaries and wrapping
12 rotary_encoder.set(min_val=0, max_val=20, range_mod
13
14 last_value = rotary_encoder.value()
15
16 while True:
17     current_value = rotary_encoder.value()
18
19     if current_value != last_value:
20         print("Counter value:", current_value)
21
22         if current_value > last_value:
```

Shell

```
Counter value: 1
Direction: decreased
Counter value: 0
Direction: decreased
Counter value: 20
Direction: increased
Counter value: 19
Direction: decreased
Counter value: 18
Direction: decreased
```

Joystick (joystick.py)

joystick.py

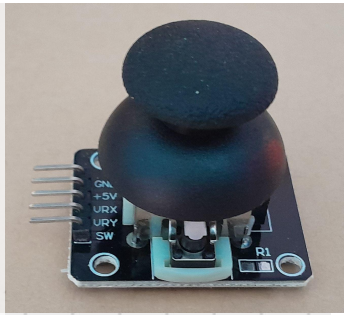
```
from DIYables_MicroPython_Joystick import Joystick
from machine import Pin, ADC
import time

VRX_PIN = 33 # connect to VRX pin of joystick
VRY_PIN = 32 # connect to VRY pin of joystick
SW_PIN = 25  # connect to SW pin of joystick

adc_vrx = ADC(Pin(VRX_PIN))
adc_vry = ADC(Pin(VRY_PIN))
# Set the ADC width (resolution) to 12 bits
adc_vrx.width(ADC.WIDTH_12BIT)
adc_vry.width(ADC.WIDTH_12BIT)
# Set the attenuation to 11 dB, allowing input range up
adc_vrx.atten(ADC.ATTN_11DB)
adc_vry.atten(ADC.ATTN_11DB)

joystick = Joystick(pin_x=VRX_PIN, pin_y=VRY_PIN, pin_
# Configure the debounce time if necessary (default is
joystick.set_debounce_time(100)  # debounce time set to
```

Shell

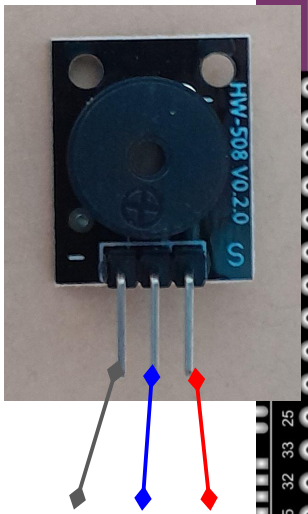
```
Joystick Position - X: 0, Y: 1927, pressed count: 4
Joystick Position - X: 0, Y: 1945, pressed count: 4
Joystick Position - X: 0, Y: 1930, pressed count: 4
Joystick Position - X: 0, Y: 1933, pressed count: 4
Joystick Position - X: 0, Y: 1931, pressed count: 4
Joystick Position - X: 0, Y: 1931, pressed count: 4
Joystick Position - X: 0, Y: 1931, pressed count: 4
Joystick Position - X: 0, Y: 1930, pressed count: 4
Joystick Position - X: 0, Y: 1922, pressed count: 4
Joystick Position - X: 0, Y: 1945, pressed count: 4
```

Joystick - ESP32

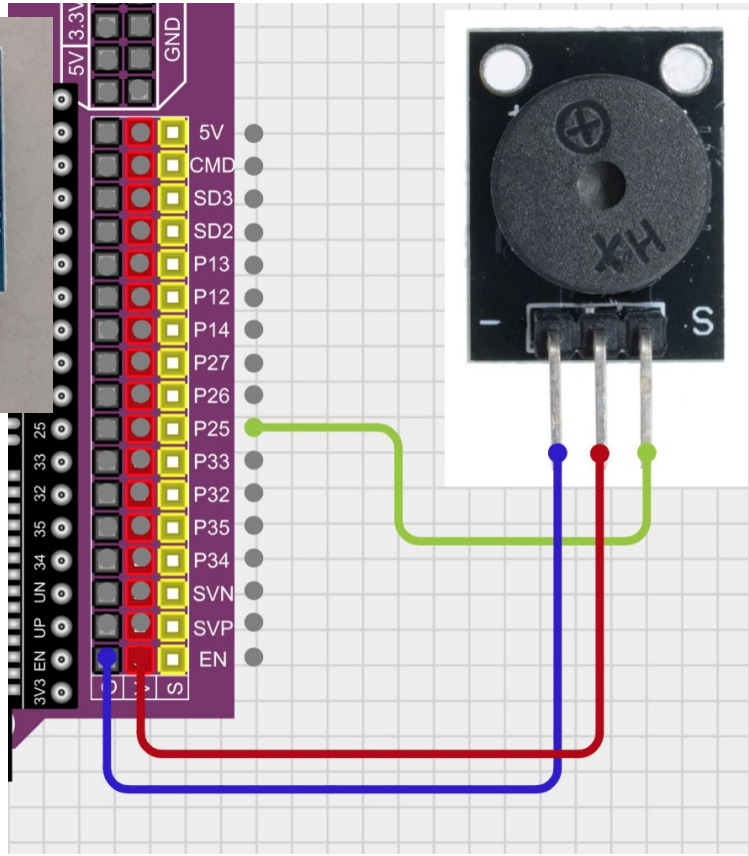
GND	GND (black)
+5V	VCC (red)
VRX	33
VRY	32
SW	25



Making simple sounds with the buzzer (buzzer.py)



GND + 25



buzzer.py

```
1  import machine
2  import time
3
4  # Define the GPIO pin that is connected to the buzzer
5  buzzer = machine.PWM(machine.Pin(25))
6
7  # Define the frequencies of the notes in Hz
8  C5 = 523
9  D5 = 587
10 E5 = 659
11 F5 = 698
12 G5 = 784
13 A5 = 880
14 B5 = 988
15
16 # Define the durations of the notes in milliseconds
17 quarter_note = 250
18 half_note = 300
19 whole_note = 1000
20
21 # Define the melody as a list of tuples (note, duratio
22 melody = [
```

Shell

```
>>> %Run -c $EDITOR_CONTENT
```

Session 1-3 prompt

Design something that turns everyday moments into a show.

If you are dry on ideas…

- A wacky music instrument control panel
- A control panel that looks or acts like a creature